Reyhane Javidi Alsaadi

Computer Engineering Scholar · Edge AI & Vision Foundation Models Researcher
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EDUCATION & SCHOLASTIC STANDING

Dr. Shariaty Technical University Tehran, Iran
Bachelor of Science in Computer Engineering **Cumulative GPA: 19.00 / 20.00** (≈**3.95 / 4.00**)
• **Academic Standing: Rank 1** in Department / Dean's Honor Roll (Summa Cum Laude equivalent).
• **Selected High-Distinction Coursework (All Top Grades):** Linear Algebra & Vector Calculus (**20.00/20**), Advanced Algorithms & Data Structures (**20.00/20**), Machine Learning & Neural Networks (**20.00/20**), Probability & Engineering Statistics (**19.50/20**), Software Architecture (**20.00/20**), Operating Systems (**19.50/20**).

PUBLICATIONS & SCIENTIFIC PREPRINTS

BioDINO-Edge: Biologically-Grounded Multi-Scale Vision Foundation Models for Trustworthy Few-Shot Plant Pathology on Resource-Constrained Hardware
Reyhane Javidi Alsaadi *(Lead Author), Mohammad Reza Samad Zadeh* *Target: IEEE T-PAMI / CVPR (2026)*
• Formulated an analytical closed-form Fisher-LDA alignment via **Ledoit-Wolf covariance shrinkage** on a joint 896-d manifold fusing **DINOv2 Small** (spatial patch tokens) and **BioCLIP ViT-B** (taxonomic hierarchies), resolving high-dimensional singularity in < 0.003 ms.
• Introduced **Purified Botanical Prompts** and **Soft-Anchored Transductive Calibration**, reaching **98.72% peak accuracy** on PlantVillage (11 novel classes) and **83.61%** on in-the-wild PlantDoc (**+13.00%** over recent 2026 multimodal SOTA).
• Packaged as a Dual-Tier Edge architecture: 22.05 MB standalone INT8 model (>45 FPS on CPU, ECE calibrated to 0.81%).

HONORS, AWARDS & COMMENDATIONS

• **University Official Letter of Commendation:** Awarded a formal letter of honor by university department leadership in recognition of sustained academic distinction, leadership in engineering initiatives, and research dedication.
• **First-Rank Scholar Distinction:** Recognized as the top-ranking undergraduate student across the Faculty of Computer Engineering throughout undergraduate tenure.

TEACHING EXPERIENCE & ACADEMIC MENTORSHIP

Department of Computer Engineering, Dr. Shariaty Technical University Tehran, Iran
Teaching Assistant (TA) — Game Development & Interactive Graphics Programming *1 Academic Term*
• Mentored over 40 undergraduate engineering students in algorithmic game architecture, state management, graphics rendering, and physics simulations.
• Conducted weekly problem-solving and code review sessions; designed laboratory grading rubrics.
• **Verification:** Supported by an **Official University Verification Letter** attesting to instructional excellence and student mentorship.

SELECTED ENGINEERING & SYSTEMS PROJECTS

BioDINO-Edge Mobile (100% Offline Android AI Scanner) *Kotlin 2.0, Compose, ONNX Mobile NNAPI*
• Native Android edge application executing foundation model inference at >**55 FPS** on ARM Cortex processors.
• Engineered a **Zero-GC direct memory pipeline** (`PreallocatedTensor`) eliminating Garbage Collector stutter; integrated a 3-layer security gate (Luminance, Foliage Manifold score ≥ 0.40, Shannon Margin-Entropy confidence) to prevent false positives on dark or non-leaf frames.

Scalable AI Backend & Microservices Architecture *NestJS, PostgreSQL, Redis, Next.js 16*
• Designed modular backend ingestion microservices in **NestJS** (TypeScript) with strict dependency injection principles.
• Modeled relational time-series foliar diagnostics in **PostgreSQL**; implemented high-throughput in-memory caching and queuing in **Redis**.
• Developed reactive frontend dashboards using **React** and **Next.js 16** under strict Git workflow discipline.

TECHNICAL & RESEARCH COMPETENCIES

• **AI & Computer Vision:** PyTorch, TorchVision, Timm, OpenCLIP, Vision Transformers (ViT), DINOv2, BioCLIP, Few-Shot Learning (Fisher-LDA, MAML), Covariance Shrinkage (Ledoit-Wolf), Self-Attention Rollout, Scikit-Learn, NumPy, SciPy.
• **Backend & Databases:** NestJS, Node.js, TypeScript, PostgreSQL (SQL, Relational Modeling), Redis (Caching & Queues), REST APIs.
• **Frontend & Web:** React, Next.js 16 (App Router, Static Export), JavaScript (ES6+), Tailwind CSS, Responsive Web Design.
• **Edge & Mobile:** Kotlin 2.0, Jetpack Compose, ONNX Runtime Mobile, Android NNAPI, INT8 Dynamic Quantization, Android NDK.
• **Developer Tools:** Git, GitHub Actions (CI/CD), Linux/Bash, LaTeX (IEEEtran, Article), Technical Research Writing.